

POWERFACTORY

PowerFactory 2024

Technical Reference

Booster Transformer

ElmTrb, TypTrb

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

Publisher:
DIgSILENT GmbH
Heinrich-Hertz-Straße 9
72810 Gomaringen / Germany
Tel.: +49 (0) 7072-9168-0
Fax: +49 (0) 7072-9168-88
info@digsilent.de

Please visit our homepage at:
https://www.digsilent.de

Copyright © 2024 DIgSILENT GmbH
All rights reserved. No part of this publication may be reproduced or distributed in any form without written permission of DIgSILENT GmbH.

March 28, 2024
PowerFactory 2024
Revision 2

Contents

1 General Description **1**

2 Three-phase model **1**

3 Single-phase model **2**

4 Model description **3**

4.1 Three-phase model 3

4.2 Single-phase model 4

4.3 Input and equivalent circuit parameters 5

4.3.1 Positive- and negative sequence 6

4.3.2 Zero sequence 6

4.3.3 Positive-, negative- and zero sequence for the EMT simulation 7

5 Harmonics **8**

5.1 K-Factor, Factor-K and FHL 8

5.1.1 K-Factor 9

5.1.2 Factor-K 9

5.1.3 FHL 9

5.1.4 Input data 9

6 Inputs/Outputs/State variables of the dynamic models **10**

6.1 EMT-Model 10

7 References **11**

List of Figures **12**

List of Tables **13**

1 General Description

*PowerFactory*comes with a built-in model for three-phase and single-phase booster transformers. The model of the booster transformer is presented in this document.

The transformer model in *PowerFactory* consists of the booster transformer element (*ElmTrb*), and the booster transformer type (*TypTrb*). The transformer element allows input of data relating to the control of the transformer under steady-state conditions, and the transformer type allows input of the physical properties of the transformer.

The *Technology* field in the type must agree with the *No. of Phases* field in the element.

Saturation can be defined in the booster transformer type and it is modelled in the same way as for the three-phase transformer model. Hysteresis is not supported for the booster transformer.

2 Three-phase model

Three-phase booster transformers are also called phase angle regulating transformers or phase-shifters.

The single-line representation of the three-phase booster transformer is shown in Figure 2.1.

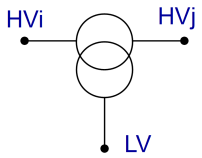


Figure 2.1: Single-line representation of the three-phase booster transformer

The three-phase booster transformer is a three-port element connecting three cubicles in the network.

The three-phase booster transformer can be used to control the flow of active and reactive power through the HV sides by modifying the voltage in magnitude and angle at the LV side. For example, this can be done by connecting a regulating transformer or power electronic equipment at the LV side.

The three phase circuit diagrams for a LV YN and a LV delta connection are shown in Figure 2.2.

The LV side neutral can be connected to the LV terminal neutral or to a separate neutral bus. The transformer LV side can be grounded via a grounding impedance. These settings are defined on the *Basic Data - Grounding/Neutral Conductor* page.

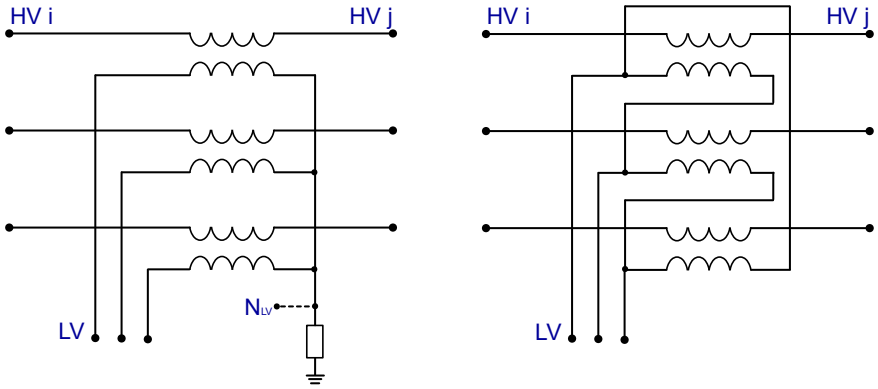


Figure 2.2: Circuit diagram of the booster transformer (left YN connection, right Δ connection)

3 Single-phase model

The single-phase booster transformer is a four-port element connecting four cubicles in the network.

The single-line representation of the three-phase booster transformer is shown in Figure 3.1.

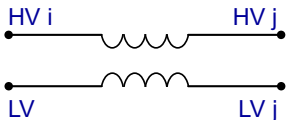


Figure 3.1: Single-line representation of the single-phase booster transformer

Note that the left side low voltage terminal is named ”LV”. Only the positive sequence data in the booster transformer type are required for the single-phase model.

4 Model description

4.1 Three-phase model

The detailed positive- and negative-sequence model with impedances in per-unit is shown in Figure 4.1.

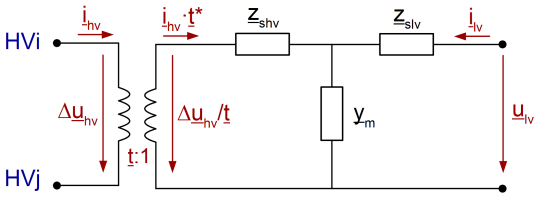


Figure 4.1: Positive-sequence equivalent circuit of the three-phase booster transformer

In Figure 4.2, the zero-sequence equivalent circuits in per-unit are shown for Y and Δ LV connection groups.

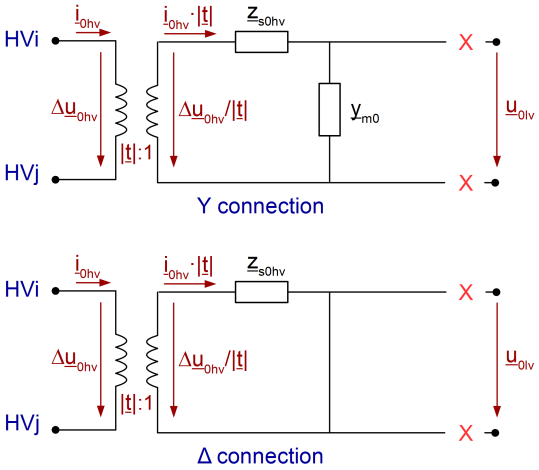


Figure 4.2: Zero-sequence equivalent circuits of the three-phase booster transformer, Y and Δ LV connection groups

In Figure 4.3, the zero-sequence equivalent circuit in per-unit is shown for YN LV connection group.

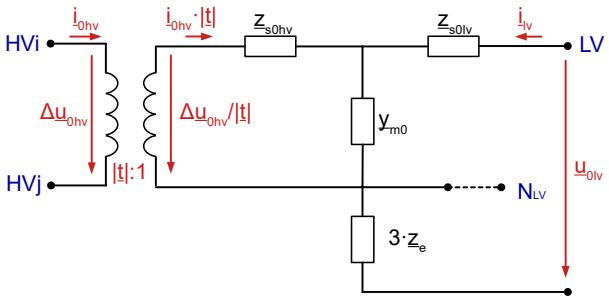


Figure 4.3: Zero-sequence equivalent circuits of the three-phase booster transformer, YN LV connection group

4.2 Single-phase model

The detailed model with impedances in per-unit is shown in Figure 4.4.

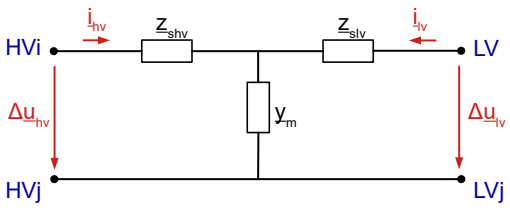


Figure 4.4: Equivalent circuit of the single-phase booster transformer

4.3 Input and equivalent circuit parameters

Using the transformer input data, the impedances used by the booster transformer model are calculated.

The nominal power of the booster transformer in *PowerFactory* is the full nominal power (input parameter $strn$) multiplied by the number of parallel booster transformers (input parameter $ntnum$):

$$S_{nom} = ntnum \cdot strn \tag{1}$$

An impedance in Ω as seen from one of the connection sides, referred to the voltage of that connection side, needs to be divided by the base impedance to get the p.u. value and then multiplied with 100 to get the value in %:

$$uktr = \frac{Z_{\Omega}}{Z_{base}} \cdot 100\ \% \tag{2}$$

where the base impedance is calculated depending on which side Z_{Ω} is measured:

$$Z_{base} = \begin{cases} \dfrac{uktr_h^2}{strn} & \Omega \quad \text{when } Z_{\Omega} \text{ is measured at the HV side} \\ \dfrac{uktr_l^2}{strn} & \Omega \quad \text{when} Z_{\Omega} \text{ is measured at the LV side} \end{cases} \tag{3}$$

where $uktr_h$ is the nominal voltage difference of the HV connection sides and $uktr_l$ is the nominal voltage of the LV winding.

The no-load current $curmg$ in %, based on the apparent power of the transformer, is calculated from the measured no-load current I_0 in kA at the LV winding as:

$$curmg = \frac{I_0}{I_{base_lv}} \cdot 100 \tag{4}$$

where

$$I_{base_lv} = \frac{strn}{\sqrt{3} \cdot utrn_l} \tag{5}$$

for the three-phase booster and

$$I_{base_lv} = \frac{strn}{utrn_l} \tag{6}$$

for the single-phase booster.

The nominal voltage difference of the HV connection sides $utrn_h$ is used when calculating the per-unit value of the HV side voltage difference (shown in Figure 4.1):

$$\Delta \underline{u}_{hv} = \frac{\underline{U}_{bus_HVi} \cdot \underline{u}_{hvi} - \underline{U}_{bus_HVj} \cdot \underline{u}_{hvj}}{utrn_h} \quad \text{p.u.} \tag{7}$$

where $\underline{U}_{bus_HVi}$ and $\underline{U}_{bus_HVj}$ are the nominal voltages of the connected terminals ($uknom$) and $\underline{u}_{HVi}$ and $\underline{u}_{HVj}$ are the actual voltage per-unit values. The negative and zero sequence voltage differences are calculated similarly.

For the three-phase booster transformer, the per-unit value of the LV voltage is calculated as:

$$\underline{u}_{lv} = \frac{\underline{U}_{bus_LV} \cdot \underline{u}_{lv}}{utrn_l} \quad \text{p.u.} \tag{8}$$

For the single-phase booster transformer, the per-unit value of the LV side voltage difference is calculated as:

$$\Delta\underline{u}_{lv} = \frac{\underline{U}_{bus_LV} \cdot \underline{u}_{lv} - \underline{U}_{bus_LVj} \cdot \underline{u}_{lvj}}{utrn_l} \quad \text{p.u.} \tag{9}$$

4.3.1 Positive- and negative sequence

Having the short-circuit voltages $uktr$ and copper losses $pcutr$, the positive sequence impedance $\underline{z}$ can be calculated and then using the distribution of the short circuit impedance $itrdz$ also the impedances of the HV and LV sides:

$$\underline{z}.r = \frac{pcutr}{1000 \cdot strn} \quad \text{p.u.} \tag{10}$$

$$\underline{z}.i = \sqrt{\frac{uktr}{100}^2 - \underline{z}.r^2} \quad \text{p.u.} \tag{11}$$

$$\underline{z}_{shv} = \underline{z} \cdot itrdz \quad \text{p.u.} \tag{12}$$

$$\underline{z}_{slv} = \underline{z} \cdot (1 - itrdz) \quad \text{p.u.} \tag{13}$$

The magnetising admittance $\underline{y_m}$ is calculated using the no-load current in % $curmg$ (no-load losses are neglected):

$$\underline{y}_m = -\jmath \cdot \frac{curmg}{100} \quad \text{p.u.} \tag{14}$$

The complex winding ratio of the LV-HV side is calculated only using the phase shift $nt2ag$ of the transformer:

$$\underline{t} = \cos\left(\frac{nt2ag \cdot \pi}{6}\right) - \jmath \cdot \sin\left(\frac{nt2ag \cdot \pi}{6}\right) \tag{15}$$

4.3.2 Zero sequence

Using the zero sequence short-circuit voltages $uk0tr$ and the resistive part $ur0tr$, the zero sequence impedance $\underline{z}_0$ can be calculated:

$$\underline{z}_0.r = \frac{ur0tr}{100} \quad \text{p.u.} \tag{16}$$

$$\underline{z}_0.i = \sqrt{\frac{uk0tr}{100}^2 - \underline{z}.r^2} \quad \text{p.u.} \tag{17}$$

The HV zero sequence impedance is calculated using the zero-seq. distribution factor as:

$$\underline{z}_{s0hv} = \underline{z}_0 \cdot zx0hl_h \quad \text{p.u.} \tag{18}$$

The LV zero sequence impedance is calculated using the zero-seq. distribution factor as:

$$\underline{z}_{s0lv} = \begin{cases} \underline{z}_0 \cdot zx0hl_l + 3 \cdot \underline{z}_{elv} & \text{p.u.} \quad \text{for YN connection} \\ 0 & \text{p.u.} \quad \text{for Y and D connections} \end{cases} \tag{19}$$

where the earthing impedance $\underline{z}_{elv}$ is calculated using the LV-side internal grounding impedances and is based on the apparent power of the transformer and the LV winding nominal voltage $utrn_l$:

$$\underline{z}_{elv} = (re0tr_l + \jmath \cdot xe0tr_l) \cdot \frac{strn}{utrn_l^2} \tag{20}$$

The zero-sequence magnetising admittance $\underline{y}_{m0}$ is calculated using the zero-sequence magnetizing reactance to $uk0tr$ ratio $zx0hl_n$:

$$\underline{y}_{m0} = \begin{cases} -\jmath \cdot \dfrac{1}{\dfrac{uk0tr}{100} \cdot zx0hl_n} & \text{p.u.} \quad \text{for Y and YN connections} \\ 0 & \text{p.u.} \quad \text{for D connections} \end{cases} \tag{21}$$

The absolute value of the complex winding ratio $|\underline{t}|$ is always 1 since it is calculated only by using the phase shift $nt2ag$ of the transformer.

4.3.3 Positive-, negative- and zero sequence for the EMT simulation

The resistances and inductances used for the EMT model are obtained from the complex impedance values calculated above as follows:

$$r = \underline{z}.r \tag{22}$$

$$l = \frac{\underline{z}.i}{2 \cdot \pi \cdot F_{nom}} \tag{23}$$

$$\tag{24}$$

where F_{nom} is the nominal frequency in Hz.

The magnetising susceptance is calculated using the no-load current $curmg$ as:

$$b_m = \frac{curmg}{100} \cdot 2 \cdot \pi \cdot f_{nom} \qquad \text{p.u.} \tag{25}$$

The zero-sequence magnetising susceptance is calculated using the zero-sequence magnetizing reactance to $uk0tr$ ratio $zx0hl_n$:

$$b_{m0} = \begin{cases} \dfrac{2 \cdot \pi \cdot f_{nom}}{\dfrac{uk0tr}{100} \cdot zx0hl_n} & \text{p.u.} \quad \text{for Y and YN connections} \\ 0 & \text{p.u.} \quad \text{for D connections} \end{cases} \tag{26}$$

5 Harmonics

For the three-phase model, it is possible to define a frequency characteristic for the following quantities: positive sequence impedance ($r1$ and $l1$), zero sequence impedance ($r0$ and $l0$) and positive sequence magnetizing admittance ($1/lm$ and gm).

For the three-phase model, it is possible to define a frequency characteristic for the following quantities: positive sequence impedance ($r1$ and $l1$) and positive sequence magnetizing admittance ($1/lm$ and gm).

5.1 K-Factor, Factor-K and FHL

Transformers experience increased losses in the presence of power system harmonic currents. In the worst-case, excessive losses can lead to transformer overheating and subsequent failure. To assist in the selection of an appropriate transformer, various factors are available:

- K-factor (UL 1562); mainly used in the US
- Factor-K (BS 7821); mainly used in Europe
- Harmonic Loss Factor (FHL) (IEEE C.57.110-1998)

These factors are indicators of the ability of a transformer to handle harmonic loads. Non-linear loads in the power system produce harmonic currents which are capable of causing unwanted side-effects, including increased transformer losses. Transformer losses are comprised of:

- Stray magnetic losses in the transformer core; and
- Eddy current and ohmic losses in the transformer windings.

In the presence of harmonic currents, eddy current losses can become large because they increase with the square of the frequency. The winding eddy current losses (due to the skin effect) at harmonic order h are given by:

$$Pe_h = Pe_f1 \cdot {I_h}^2 \cdot h^2 \tag{27}$$

where Pe_f1 represents the rated eddy current losses (based solely on data from the transformer type); Pe_h is the eddy current loss at harmonic order h; and I_h is the fraction of total rms load current at harmonic order h. The calculation variable *Po_f1* represents the rated ohmic losses (also based solely on data from the transformer type).

In terms of *PowerFactory* transformer type data, the rated eddy current losses, *Pe_f1*, are calculated in kW as follows:

$$Pe_f1 = (zshv{:}r + zslv{:}r) \cdot strn \cdot 1000 \cdot eddypc \tag{28}$$

The total eddy current loss, Pe_tot, is given by the following summation:

$$Pe_tot = Pe_f1 \cdot \sum_{h=1}^{h_{max}} {I_h}^2 \cdot h^2 \tag{29}$$

5.1.1 K-Factor

The K-Factor accounts for the increased eddy current losses due to harmonic currents. Mathematically, it is the ratio of eddy current losses in the presence of non-linear and linear loads [2]:

$$K = \frac{Pe_tot}{Pe_f1} = \sum_{h=1}^{h_{max}} {I_h}^2 \cdot h^2 \tag{30}$$

Following the calculation of the K-Factor, an appropriate K-transformer can then be selected which has a higher K-rating.

5.1.2 Factor-K

The factor-K was introduced in [1] and is described mathematically by:

$$K = \left[1 + \frac{e}{1+e}\left(\frac{I_1}{I}\right)^2 \cdot \sum_{h=2}^{h_{max}} \left(h^q \left(\frac{I_h}{I_1}\right)^2\right)\right]^{0.5} \tag{31}$$

where e is the eddy current loss at the fundamental frequency divided by the loss due to a dc current equal to the rms values of the sinusoidal current, both at reference temperature. The harmonic order is represented by h, and the exponential constant, q, depends on the type of winding and the frequency. Typical values are 1.7 for transformers utilising round/rectangular cross-section conductors in both windings, and 1.5 for transformers which use foil-type low voltage windings. This value should be available from the transformer manufacturer. I is the rms value of the sinusoidal current including all harmonics, and is given by:

$$I = \left(\sum_{h=1}^{h_{max}} (I_h)^2\right)^{0.5} = I_1 \left[\sum_{h=1}^{h_{max}} \left(\frac{I_h}{I_1}\right)^2\right]^{0.5} \tag{32}$$

5.1.3 FHL

The FHL is described mathematically by [2]:

$$FHL = \frac{\sum_{h=1}^{h_{max}} \left[\frac{I_h}{I_1}\right]^2 \cdot h^2}{\sum_{h=1}^{h_{max}} \left[\frac{I_h}{I_1}\right]^2} \tag{33}$$

5.1.4 Input data

For the calculation of any of these factors, the ratio of winding eddy current losses to copper losses should be entered in the transformer type (*Harmonics/Power Quality* page) using input parameter *Ratio: winding eddy current-/copper losses*.

Additionally, for the calculation of Factor-K, the exponent q (from (31)) must be entered in the *Harmonic Load Flow* command (*ComHldf*) via input parameter *Calculation of Factor-K for Transformers (Exponent)*.

By default, the value of the input parameter *Ratio: winding eddy current-/copper losses* is set to 0.

6 Inputs/Outputs/State variables of the dynamic models

6.1 EMT-Model

Table 6.1: State variables definition of the three-phase EMT-model

Parameter	Description	Unit
psimd	Magnetising Flux, alpha component	p.u.
psimq	Magnetising Flux, beta component	p.u.
psim0	Magnetising Flux, gamma component	p.u.

Table 6.2: State variables definition of the single-phase EMT-model

Parameter	Description	Unit
psim	Magnetising Flux	p.u.

7 References

[1] Three phase oil-immersed distribution transformers, 50 hz, from 50 to 2500 kva with highest voltage for equipment not exceeding 36 kv. Determination of the power rating of a transformer loaded with non-sinusoidal currents, 1995.

[2] IEEE Std c57.110-1998 Recommended Practice for Establishing Transformer Capability, 1999.

List of Figures

2.1 Single-line representation of the three-phase booster transformer 1

2.2 Circuit diagram of the booster transformer (left YN connection, right Δ connection) 2

3.1 Single-line representation of the single-phase booster transformer 2

4.1 Positive-sequence equivalent circuit of the three-phase booster transformer . . . 3

4.2 Zero-sequence equivalent circuits of the three-phase booster transformer, Y and Δ LV connection groups 3

4.3 Zero-sequence equivalent circuits of the three-phase booster transformer, YN LV connection group 4

4.4 Equivalent circuit of the single-phase booster transformer 4

List of Tables

6.1 State variables definition of the three-phase EMT-model 10

6.2 State variables definition of the single-phase EMT-model 10